WORKSHEET 2 : DOUBLE AND TRIPLE INTEGRALS

1 Double Integrals

Exercise 1 (session 1)

For each of the following functions, sketch the domain D and calculate the double integral $\iint\limits_D f(x,y)\,dx\,dy$.

1. $f(x,y) = x$ *and* $D = \{(x,y) \in \mathbb{R}^2\,;\, y \geq 0,\, x - y + 1 \geq 0,\, x + 2y - 4 \leq 0\}$.
2. $f(x,y) = x + y$ *and* $D = \{(x,y) \in \mathbb{R}^2\,;\, 0 \leq x \leq 1,\, x^2 \leq y \leq x\}$.
3. $f(x,y) = \cos(xy)$ *and* $D = \left\{(x,y) \in \mathbb{R}^2\,;\, 1 \leq x \leq 2,\, 0 \leq xy \leq \frac{\pi}{2}\right\}$.
4. $f(x,y) = \frac{1}{(x+y)^3}$ *and* $D = \{(x,y) \in \mathbb{R}^2\,;\, 1 < x < 3,\, y > 2,\, x + y < 5\}$.

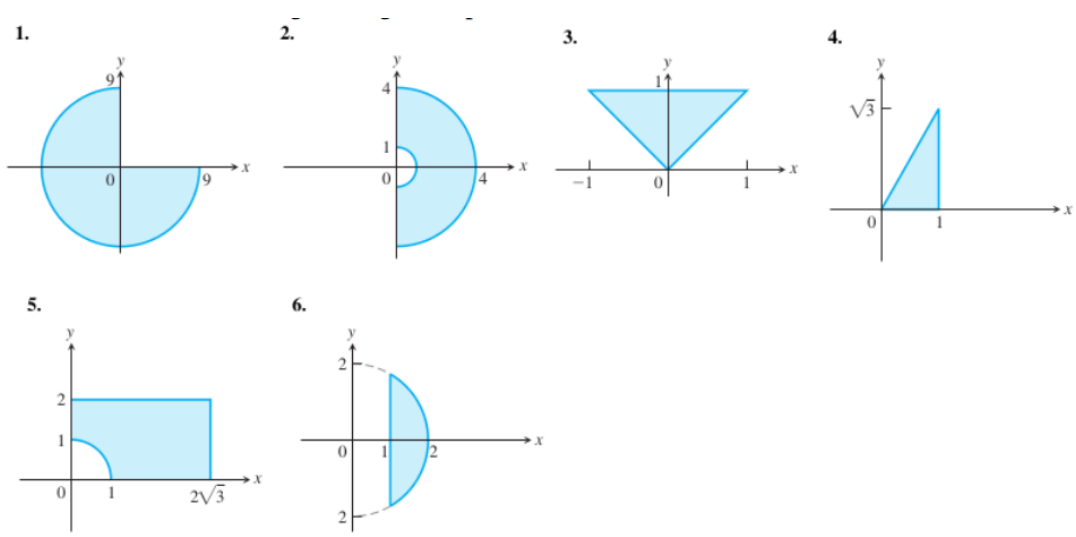
Exercise 2 (session 1)

Geometrically describe the integration domain $D \subset \mathbb{R}^2$ of the following double integrals, then change the order of integration.

$$\int_0^1 \int_{y^2}^{\sqrt{y}} f(x,y)\,dx\,dy.$$

Exercise 3 (session 1)

Describe the given region in polar coordinates.



7. *The region enclosed by the circle* $x^2 + y^2 = 2x$.
8. *The region enclosed by the semicircle* $x^2 + y^2 = 2y,\ y \geq 0$.

Exercise 4 (session 1)

Using polar coordinates, calculate the integrals $\iint\limits_D f(x,y)\,dx\,dy$ where :

1. *D is the annulus bounded by circles centered at O with radii $0 < a < b$ and $f(x,y) = \frac{1}{x^2+y^2}$.*
2. *D is the disk centered at O with radius a and $f(x,y) = (x+y)^2$.*
3. *D is bounded by the circle centered at O with radius 3 and the circle centered at $(1,0)$ with radius 1, and $f(x,y) = x^2 + y^2$.*
4. *D is the set of points in the disk centered at 0 with radius 1 such that $0 \leq y \leq x$ and $f(x,y) = (x-y)^2$.*
5. *D is the set of points in the square $[0,1] \times [0,1]$ outside the circle centered at O with radius 1, and $f(x,y) = \frac{xy}{1+x^2+y^2}$.*

Exercise 5 (session 2)

Change the Cartesian integral into an equivalent polar integral. Then evaluate the polar integral.

1. $\displaystyle\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}} dy\,dx$
2. $\displaystyle\int_{0}^{1}\int_{0}^{\sqrt{1-y^2}} \sqrt{x^2+y^2}\,dx\,dy$
3. $\displaystyle\int_{0}^{2}\int_{0}^{\sqrt{4-y^2}} \sqrt{x^2+y^2}\,dx\,dy$
4. $\displaystyle\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dy\,dx$
5. $\displaystyle\int_{0}^{6}\int_{0}^{y} x\,dx\,dy$
6. $\displaystyle\int_{1}^{2}\int_{0}^{\sqrt{2x-x^2}} \frac{1}{(x^2+y^2)^2}\,dy\,dx$
7. $\displaystyle\int_{0}^{2}\int_{0}^{x} y\,dy\,dx$
8. $\displaystyle\int_{0}^{1}\int_{x}^{\sqrt{2-x^2}} (x+2y)\,dy\,dx$
9. $\displaystyle\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \ln(x^2+y^2+1)\,dx\,dy$

Exercise 6 (session 2)

Using changes of variables, calculate the following double integrals :

1. $\displaystyle\iint_D e^{\left(\frac{x-y}{x+y}\right)}\,dx\,dy$ *where* $D = \{(x,y) \in \mathbb{R}^2 : x+y \leq 1,\ x > 0,\ y > 1\}$
   (hint : let $u = x - y$ and $v = x + y$)
2. $\displaystyle\iint_D \frac{y}{x}\,dx\,dy$ *where* $D = \{(x,y) \in \mathbb{R}^2 : x \leq y \leq 2x,\ x \leq y^2 \leq 2x\}$
   (hint : let $u = \frac{x}{y}$ and $v = \frac{y^2}{x}$)
3. $\displaystyle\iint_D \frac{y^2(2+2y^2-x)}{(2-x)^4}\,dx\,dy$ *where* $D = \{(x,y) \in \mathbb{R}^2 : x-1 < y^2 < x,\ 0 < y < 2-x\}$
   (hint : let $u = \frac{y}{(2-x)}$ and $v = y^2 - x$)

Exercise 7 (session 2)

Let $D = \{(x,y) \in \mathbb{R}^2 \,|\, x > 0,\ 1 \leq xy \leq 2,\ 1 \leq x^2 - y^2 \leq 4\}$.

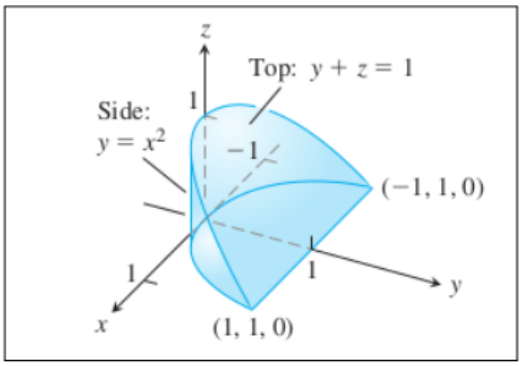
1. *Sketch the domain D.*
2. *Let the mapping $\Phi :]0,+\infty[^2 \to]0,+\infty[^2$ defined by $(x,y) \mapsto (xy, x^2-y^2)$.*
   Find $\Phi(D) = \Delta$ explicitly.
3. *Calculate* $\displaystyle\iint_D \frac{xy(x^2+y^2)}{x^2-y^2}\,dx\,dy$.

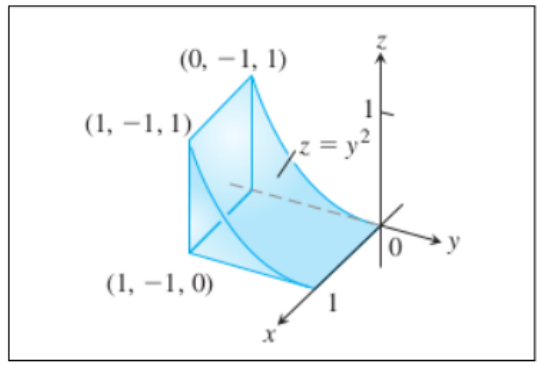
2 Triple Integrals

Exercise 8 **(session 3)**

Here are the regions of integration of the integrals

$$I = \int_{-1}^{1}\int_{x^2}^{1}\int_{0}^{1-y} dz\,dy\,dx \quad \text{and} \quad J = \int_{0}^{1}\int_{-1}^{0}\int_{0}^{y^2} dz\,dy\,dx$$





Rewrite the integrals I and J as equivalent iterated integrals in the order :

- $dy\,dz\,dx$, • $dy\,dx\,dz$, • $dx\,dy\,dz$, • $dx\,dz\,dy$, • $dz\,dx\,dy$

Exercise 9 **(session 3)**

Calculate : $\displaystyle\iiint_D f(x,y,z)\,dx\,dy\,dz$ *for*

1. $f(x,y,z) = \cos x$ *and* $D = \{(x,y,z) \in \mathbb{R}^3\,;\, x^2+y^2+z^2 < 1\}$.
2. $f(x,y,z) = xyz$ *and* $D = \{(x,y,z) \in \mathbb{R}^3\,;\, 0 \le x \le y \le z \le 1\}$.
3. $D = \{(x,y,z)\,|\,4 \le x^2+y^2+z^2 \le 16\}$ *and* $f(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$
4. $D = \{(x,y,z)\,|\,x^2+y^2+z^2 \le 1,\ y^2-2xz \le 0,\ x^2+y^2 \ge 4z^2,\ z \ge 0\}$ *and* $f(x,y,z) = \frac{\sqrt{z}}{\sqrt{x^2+y^2}}$

Exercise 10 **(session 3)**

Sketch the domains described algebraically and calculate the following triple integrals :

1. $V_1 = \{(x,y,z) \in \mathbb{R}^3,\ x>0,\ y>0,\ 0 \le z \le 1-y^2,\ x+y \le 1\}$, $I_1 = \displaystyle\iiint_{V_1} z\,dx\,dy\,dz$
2. $V_2 = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2 \le 3z,\ x^2+y^2+z^2 \le 4\}$, $I_2 = \displaystyle\iiint_{V_2} dx\,dy\,dz$
3. $V_3 = \{(x,y,z) \in \mathbb{R}^3,\ x>0,\ y>0,\ z>0,\ x+y+z \le 1\}$, $I_3 = \displaystyle\iiint_{V_3} \frac{z\,dx\,dy\,dz}{(1+x+y+z)^3}$
4. $V_4 = \{(x,y,z) \in \mathbb{R}^3,\ 0 \le z \le 4-4x^2-y^2\}$, $I_4 = \displaystyle\iiint_{V_4} \sqrt{x^2+y^2}\,z\,dx\,dy\,dz$

Exercise 11 **(session 4)**

By performing changes of variables, calculate the following integrals over the described domains :

1. $I = \displaystyle\iiint_V \sqrt{x^2+y^2}\,dx\,dy\,dz$ *where* $V = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2 \le (z-1)^2,\ 0 \le z \le 1\}$

2. $I = \iiint\limits_V \frac{dx\,dy\,dz}{(x^2+y^2+1)^2}$ *where* $V = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2+z^2 \le R^2,\ x^2+y^2+z^2 \le 2Rz\}$

3. $I = \iiint\limits_V (z+2)\sqrt{x^2+y^2+1}\,dx\,dy\,dz$ *where* $V = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2 \le z^2,\ 0 \le z \le 2h\}$

4. $I = \iiint\limits_V z\cos(x^2+y^2)\,dx\,dy\,dz$ *where* $V = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2+z^2 \le 1,\ z > 0\}$

5. *Evaluate* $\int\limits_{-3}^{3} \int\limits_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int\limits_{\sqrt{x^2+y^2}}^{3} \frac{e^z}{\sqrt{x^2+y^2}}\,dz\,dy\,dx$ *by using cylindrical coordinates.*

6. *Evaluate* $\int\limits_{-1}^{1} \int\limits_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \int\limits_{0}^{4-x^2-y^2} e^{x^2+y^2+z}\,dz\,dx\,dy$ *by using cylindrical coordinates.*

7. *Evaluate* $\iiint\limits_B \frac{dV}{\sqrt{x^2+y^2+z^2}+3}$, *where* B *is the ball of radius 2 centered at the origin.*

8. *Determine* $\iiint\limits_W \sqrt{x^2+y^2+2z^2}\,dV$ *where* W *is the solid cylinder defined by the inequalities* $x^2 + y^2 \le 4,\ -1 \le z \le 2$.

9. *Determine the value of* $\iiint\limits_W \frac{\sqrt{z}}{\sqrt{x^2+y^2}}\,dV$, *where* W *is the solid region bounded by the plane* $z = 12$ *and the paraboloid* $z = 2x^2 + 2y^2 - 6$.

10. *Find the volume of the region* W *bounded on top by* $z = \sqrt{a^2 - x^2 - y^2}$, *on the bottom by the* xy*-plane, and on the sides by the cylinder* $x^2 + y^2 = b^2$, *where* $0 < b < a$.

11. *Determine the values of the given integrals, where* W *is the region bounded by the two spheres* $x^2+y^2+z^2 = a^2$ *and* $x^2+y^2+z^2 = b^2$, *for* $0 < a < b$.

$$\iiint\limits_W \frac{dV}{\sqrt{x^2+y^2+z^2}} \qquad \iiint\limits_W \sqrt{x^2+y^2+z^2}\,e^{x^2+y^2+z^2}\,dV$$

12. *Let* W *denote the solid region in the first octant between the spheres* $x^2+y^2+z^2 = a^2$ *and* $x^2 + y^2 + z^2 = b^2$, *where* $0 < a < b$. *Determine the value of* $\iiint\limits_W (x+y+z)\,dV$

13. *Determine the value of* $\iiint\limits_W z^2\,dV$, *where* W *is the solid region lying above the cone* $z = \sqrt{3x^2+3y^2}$ *and inside the sphere* $x^2+y^2+z^2 = 6z$.

Exercise 12 (supplementary)

1. *Calculate the volume in the first octant located between the planes* $z = 0$, $z = x + y + 2$ *and inside the cylinder* $x^2 + y^2 = 16$.

2. *Calculate the volume bounded above by the paraboloid* $x^2 + 4y^2 = z$, *below by the plane* $z = 0$, *and laterally by the cylinders* $y^2 = x$ *and* $x^2 = y$.

3. *Calculate the volume bounded by the paraboloid* $x^2 + y^2 = 4z$, *the cylinder* $x^2 + y^2 = 8y$, *and the plane* $z = 0$.

4. *Calculate the missing volume when drilling a circular hole, centered at the origin, of radius $a > 0$ through a sphere of radius $2a$ centered at the origin.*

Exercise 13 (supplementary)

1. *Calculate the volume of the domain bounded by the exterior of the cone with equation $x^2 + y^2 = z^2$ and the interior of the sphere centered at the origin with radius $r > 0$.*
2. *Let $a > 0$. Calculate the volume of the domain bounded by the interior of the cylinder with equation $x^2 + y^2 = a^2$ and the interior of the sphere centered at the origin with radius $2a$.*
3. *Let $a > 0$. Calculate the common volume of the two cylinders with equations $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$.*
4. *Calculate the volume of the domain bounded by the interior of the cylinder with equation $(x-1)^2 + y^2 = 1$ and the interior of the sphere centered at the origin with radius 2.*

Exercise 14 (supplementary)

Integrate the given function over the indicated region W.

1. $f(x, y, z) = 2x - y + z$; *W is the region bounded by the cylinder $z = y^2$, the xy-plane, and the planes $x = 0$, $x = 1$, $y = -2$, $y = 2$.*
2. $f(x, y, z) = y$; *W is the region bounded by the plane $x + y + z = 2$, the cylinder $x^2 + z^2 = 1$, and $y = 0$.*
3. $f(x, y, z) = 8xyz$; *W is the region bounded by the cylinder $y = x^2$, the plane $y + z = 9$, and the xy-plane.*
4. $f(x, y, z) = z$; *W is the region in the first octant bounded by the cylinder $y^2 + z^2 = 9$ and the planes $y = x$, $x = 0$, and $z = 0$.*
5. $f(x, y, z) = 1 - z^2$; *W is the tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 3)$.*
6. $f(x, y, z) = 3x$; *W is the region in the first octant bounded by $z = x^2 + y^2$, $x = 0$, $y = 0$, and $z = 4$.*
7. $f(x, y, z) = x + y$; *W is the region bounded by the cylinder $x^2 + 3z^2 = 9$ and the planes $y = 0$, $x + y = 3$.*
8. $f(x, y, z) = z$; *W is the region bounded by $z = 0$, $x^2 + 4y^2 = 4$, and $z = x + 2$.*
9. $f(x, y, z) = 4x + y$; *W is the region bounded by $x = y^2$, $y = z$, $x = y$, and $z = 0$.*
10. $f(x, y, z) = x$; *W is the region in the first octant bounded by $z = x^2 + 2y^2$, $z = 6 - x^2 - y^2$, $x = 0$, and $y = 0$.*